6.3. Grokking Analysis

Table 5 | **Evaluation of automated AI Scientist paper generation for Grokking.**

	Total Ideas	Novel Ideas	Experiments Passed	Completed Papers	Mean Score	Max Score	Total Cost
Sonnet 3.5	51	47	25	25	3.44	5.0	~$250
GPT-4o	51	51	22	13	2.92	3.0	~$300
DeepSeek Coder	51	46	38	36	3.13	4.0	~$10
Llama-3.1 405b	51	36	30	30	2.00	3.0	~$120

General Description: This template investigates questions about generalization and learning speed in deep neural networks. We follow the classic experimental paradigm reported in Power et al. (2022) for analyzing "grokking", a poorly understood phenomenon in which validation accuracy dramatically improves long after the train loss saturates. We provide code that generates synthetic datasets of modular arithmetic tasks and then trains a Transformer model on them. Unlike the previous templates, this one is more amenable to open-ended empirical analysis (e.g. what conditions grokking occurs) rather than just trying to improve performance metrics.

Code Template: We base our implementation off of two popular open source re-implementations (May, 2022; Snell, 2021) of Power et al. (2022). The code generates four synthetic datasets of modular arithmetic tasks and trains a transformer on each across three random seeds. It returns train losses, validation losses, and the number of update steps required to reach perfect validation accuracy. The plotting scripts visualize the training and validation curves by default.

Highlighted Generated Paper 1: Unlocking Grokking: A Comparative Study of Weight Initialization Strategies in Transformer Models. This paper investigates different weight initializations and their impact on grokking. It finds that Xavier (Glorot and Bengio, 2010) and Orthogonal weight initializations consistently result in significantly faster grokking on the tasks than the widely-used default baseline weight initializations (Kaiming Uniform and Kaiming Normal). While this is a basic investigation, it provides an interesting result that could be studied in more depth. The paper also has a creative and catchy title.

Highlighted Generated Paper 2: Grokking Accelerated: Layer-wise Learning Rates for Transformer Generalization. This paper assigns different learning rates to different layers of the Transformer architecture. It finds that increasing the learning rate for higher layers results in significantly faster and more consistent grokking after iterating through different configurations throughout its experiments. It impressively includes the key section of its implementation in the write-up.

Highlighted Generated Paper 3: Grokking Through Compression: Unveiling Sudden Generalization via Minimal Description Length. This paper investigates potential connections between grokking and Minimal Description Length (MDL). We believe this idea is particularly interesting, though not executed very well. Its method for measuring MDL simply involves counting the number of parameters above a threshold ϵ. While this does end up correlating with grokking, it is not analyzed in much depth. The paper could be significantly improved by investigating other estimates of MDL and including basic ablations. Furthermore, THE AI SCIENTIST failed to write the Related Works section and hallucinated a plot (Figure 5).

Highlighted Generated Paper 4: Accelerating Mathematical Insight: Boosting Grokking Through Strategic Data Augmentation. This paper investigates data augmentation techniques for grokking in modular arithmetic. It comes up with valid and creative augmentation techniques (operand reversal and operand negation) and finds that they can significantly accelerate grokking. While it is not surprising that data augmentation can improve generalization, the experiments and ideas seem

generally well-executed. However, THE AI SCIENTIST once again failed to write the Related Works section. In principle, this failure may be easily remedied by simply running the paper write-up step multiple times.

7. Related Work

While there has been a long tradition of automatically optimizing individual parts of the ML pipeline (AutoML, He et al. (2021); Hutter et al. (2019)), none come close to the full automation of the entire research process, particularly in communicating obtained scientific insights in an interpretable and general format.

LLMs for Machine Learning Research. Most closely related to our work are those that use LLMs to assist machine learning research. Huang et al. (2024) propose a benchmark for measuring how successfully LLMs can write code to solve a variety of machine learning tasks. Lu et al. (2024a) use LLMs to propose, implement, and evaluate new state-of-the-art algorithms for preference optimization. Liang et al. (2024) use LLMs to provide feedback on research papers and find that they provide similar feedback to human reviewers, while Girotra et al. (2023) find that LLMs can consistently produce higher quality ideas for innovation than humans. Baek et al. (2024); Wang et al. (2024b) use LLMs to propose research ideas based on scientific literature search but do not execute them. Wang et al. (2024c) automatically writes surveys based on an extensive literature search. Our work can be seen as the synthesis of all these distinct threads, resulting in a single autonomous open-ended system that can execute the entire machine learning research process.

LLMs for Structured Exploration. Because LLMs contain many human-relevant priors, they are commonly used as a tool to explore large search spaces. For example, recent works have used LLM coding capabilities to explore reward functions (Ma et al., 2023; Yu et al., 2023), virtual robotic design (Lehman et al., 2023), environment design (Faldor et al., 2024), and neural architecture search (Chen et al., 2024a). LLMs can also act as evaluators (Zheng et al., 2024) for "interestingness" (Lu et al., 2024b; Zhang et al., 2024) and as recombination operators for black-box optimization with Evolution Strategies (Lange et al., 2024; Song et al., 2024) and for Quality-Diversity approaches (Bradley et al., 2024; Ding et al., 2024; Lim et al., 2024). Our work combines many of these notions, including that our LLM Reviewer judges papers on novelty and interestingness, and that many proposed ideas are new combinations of previous ones.

AI for Scientific Discovery. There has been a long tradition of AI assisting scientific discovery (Langley, 1987, 2024) across many other fields. For example, AI has been used for chemistry (Buchanan and Feigenbaum, 1981), synthetic biology (Hayes et al., 2024; Jumper et al., 2021), materials discovery (Merchant et al., 2023; Pyzer-Knapp et al., 2022; Szymanski et al., 2023), mathematics (Lenat, 1977; Lenat and Brown, 1984; Romera-Paredes et al., 2024), and algorithm search (Fawzi et al., 2022). Other works aim to analyze existing pre-collected datasets and find novel insights (Falkenhainer and Michalski, 1986; Ifargan et al., 2024; Langley, 1987; Majumder et al., 2024; Nordhausen and Langley, 1990; Yang et al., 2024; Zytkow, 1996). Unlike our work, these are usually restricted to a well-defined search space in a single domain and do not involve "ideation", writing, or peer review from the AI system. In its current form, THE AI SCIENTIST excels at conducting research ideas implemented via code; with future advances (e.g. robotic automation for wet labs (Arnold, 2022; Kehoe et al., 2015; Sparkes et al., 2010; Zucchelli et al., 2021)), the transformative benefits of our approach could reach across all science, especially as foundation models continue to improve.

8. Limitations & Ethical Considerations

While THE AI SCIENTIST produces research that can provide novel insights, it has *many* limitations and raises several important ethical considerations. We believe future versions of THE AI SCIENTIST

will be able to address many of its current shortcomings.

Limitations of the Automated Reviewer. While the automated reviewer shows promising initial results, there are several potential areas for improvement. The dataset used, from ICLR 2022, is old enough to potentially appear in the base model pre-training data - this is a hard claim to test in practice since typical publicly available LLMs do not share their training data. However, preliminary analysis showed that LLMs were far from being able to reproduce old reviews exactly from initial segments, which suggests they have not memorized this data. Furthermore, the rejected papers in our dataset used the original submission file, whereas for the accepted papers only the final camera-ready copies were available on OpenReview. Future iterations could use more recent submissions (e.g. from TMLR) for evaluation. Unlike standard reviewers, the automated reviewer is unable to ask questions to the authors in a rebuttal phase, although this could readily be incorporated into our framework. Finally, since it does not currently use any vision capabilities, The AI Scientist (including the reviewer) is unable to view figures and must rely on textual descriptions of them.

Common Failure Modes. The AI Scientist, in its current form, has several shortcomings in addition to those already identified in Section 5. These also include, but are not limited to:

- The idea generation process often results in very similar ideas across different runs and even models. It may be possible to overcome this by allowing The AI Scientist to directly follow up and go deeper on its best ideas, or by providing it content from recently-published papers as a source of novelty.
- As shown in Tables 3 to 5, Aider fails to implement a significant fraction of the proposed ideas. Furthermore, GPT-4o in particular frequently fails to write LaTeX that compiles. While The AI Scientist can come up with creative and promising ideas, they are often too challenging for it to implement.
- The AI Scientist may *incorrectly* implement an idea, which can be difficult to catch. An adversarial code-checking reviewer may partially address this. As-is, one should manually check the implementation before trusting the reported results.
- Because of The AI Scientist's limited number of experiments per idea, the results often do not meet the expected rigor and depth of a standard ML conference paper. Furthermore, due to the limited number of experiments we could afford to give it, it is difficult for The AI Scientist to conduct fair experiments that control for the number of parameters, FLOPs, or runtime. This often leads to deceptive or inaccurate conclusions. We expect that these issues will be mitigated as the cost of compute and foundation models continues to drop.
- Since we do not currently use the vision capabilities of foundation models, it is unable to fix visual issues with the paper or read plots. For example, the generated plots are sometimes unreadable, tables sometimes exceed the width of the page, and the page layout (including the overall visual appearance of the paper (Huang, 2018)) is often suboptimal. Future versions with vision and other modalities should fix this.
- When writing, The AI Scientist sometimes struggles to find and cite the most relevant papers. It also commonly fails to correctly reference figures in LaTeX, and sometimes even hallucinates invalid file paths.
- Importantly, The AI Scientist occasionally makes critical errors when writing and evaluating results. For example, it struggles to compare the magnitude of two numbers, which is a known pathology with LLMs. Furthermore, when it changes a metric (e.g. the loss function), it sometimes does not take this into account when comparing it to the baseline. To partially address this, we make sure all experimental results are reproducible, storing copies of all files when they are executed.
- Rarely, The AI Scientist can hallucinate entire results. For example, an early version of our writing prompt told it to always include confidence intervals and ablation studies. Due